



Handled by, phone
Sven Radhe, +46 8 555 521 12
E-mail
sven.radhe@sis.se

Minutes from the ninth Plenary meeting of ISO/TC 12, Quantities and units, in Prague, CZ, on 12 and 13 April 2010

Attending the meeting (Heads of delegations underlined)

Chairman	Anders J Thor (SE)
Secretary	Sven Radhe (SE)
ISO/CS	Jenny Pellaux
AFNOR	<u>Jean Schwob</u>
ANSI	<u>Ambler Thompson</u>
DIN	Burghardt Brinkmann Erik Jacobson <u>Klaus-Dieter Sommer</u>
SIS	Ann Fylkner <u>Göran Grimvall</u>
SNV	Stéphane Culatti <u>Paul Gérôme</u>
UMNZ	<u>Jan Obrdžálek</u>
ISO/TC 215	Christof Gessner
IEC/TC 25	Susana Losada Gonzalez

Apologies

Germany	Rudolf Borges
IUPAP	Leslie Pendrill

1 Opening of the meeting

The Chairman opened the meeting at 10:00 on Monday 2010-04-12. Mr. Jan Obrdžálek welcomed the delegates to Prague and UNMZ.

2 Roll call of delegates

The delegates introduced themselves (see list of participants above).

3 Adoption of the agenda

The draft agenda, ISO/TC 12, Doc. N 854, was adopted.

4 Appointment of the drafting committee

Mr. Burghardt Brinkmann, Mr. Stéphane Cullati, Mr. Jean Schwob, and Mr. Ambler Thompson were appointed to form the drafting committee. Mr. Thompson was appointed to chair the drafting committee (Resolution 1).

5 Report of the Secretariat

The Chairman, Mr. Anders J Thor, presented the report of the activities of ISO/TC 12 since the previous meeting in Copenhagen, DK, 1978, doc. ISO/TC 12 N 849. The report was approved as presented.

6 Business Plan for ISO/TC 12

The secretariat had not received any written comments to the revised draft business plan, doc. ISO/TC 12 N 853. The revision consists of the change from ISO 31 to ISO 80000 and minor changes in the template for business plans since the previous edition.

The business plan was approved and ISO/TC 12 asked ISO/CS to replace the old business plan by the new one on www.iso.org (Resolution 2).

7 Status of items of the programme of work and action to be taken

7.1 ISO 80000 and IEC 80000, *Quantities and units*, 14 parts

The update of all parts of ISO 80000 and IEC 80000 series was discussed.

Mr. Grimvall raised the question about the use of the definite article in the English versions of the documents.

Mr. Thompson asked Ms. Pellaux about the time frame required for updates and she replied that if the changes are not of a technical nature the revised documents could go directly to FDIS within ISO. The corresponding IEC procedure should be examined.

Mr. Thompson raised the issue of whether the ISO/TC 12 standards should give Unicode and hexadecimal codes. ISO/TC 12 decided that this should be the case in the revised standards (Resolution 3).

The title for ISO 80000-2 was discussed. The present title *Mathematical signs and symbols for use in the natural sciences and technology* was considered to be too long. There were two proposals; *Mathematics* and *Mathematical signs and symbols*. The latter was approved by majority vote. (Resolution 6)

DIN had made comments to the systematic review of ISO 80000-3, *Space and time*, presented in document N 845 regarding quantities used in kinematics of special relativity. The corresponding comments are also relevant for ISO 80000-4, *Mechanics* and IEC 80000-6, *Electromagnetism*. ISO/TC 12 decided to make these additions at the revision of the standards (Resolution 5). ISO/TC 12 should make IEC/TC 25 aware of the additions so that they could be coordinated with the current work on the revision of the IEC 80000-6.

Mr. Obrdžálek proposed that the title for ISO 80000-7, *Light*, should be changed to *Optics* or *Light and related electromagnetic phenomena*. The titles of the different parts of the standard are not consistent; some reflect the name of a discipline, e.g. *Mechanics* and *Acoustics*; whereas others reflect the name of a phenomenon, e.g. *Light* and *Space and time*. ISO/TC 12 decided by majority vote to keep the title of ISO 80000-7 as it is.

As consequence of this decision it was proposed that ISO 80000-8, *Acoustics* be changed to *Sound*. ISO/TC 12 decided by majority vote to make this change (Resolution 7).

ISO/TC 12 also asked its WG 3 to investigate the definition of item 8-22, sound pressure level, (Resolution 8).

There had been a comment from BIPM and IUPAC that metastable isotopes should be denoted with an "m" instead of an asterisk in ISO 80000-9, *Physical chemistry and molecular physics*, Annex B, Symbols for chemical elements and nuclides. ISO/TC 12 decided that this change should be made when the standard is revised (Resolution 9).

Mr. Jacobson asked why only some characteristic numbers are included in ISO 80000-11, *Characteristic numbers*. He has compiled a list of some 200 more numbers. ISO/TC 12 decided to ask its WG 16 to investigate the current usage of those presently in the standard, consider additions based on the input from Mr. Jacobson (Resolution 10).

Mr. Jacobson proposed that the background information for information science in IEC 80000-13, *Information science and technology*, should be improved, see also ISO 2382, *Information processing systems - Vocabulary*. ISO/TC 12 decided that this could be done when revising the standard (Resolution 11).

IEC/TC 25 decided at its plenary meeting in 2009 that IEC 80000-14, *Telebiometrics related to human physiology*, should be replaced by IEC 80003-1 (Resolution 12).

7.2 ISO 80003 and IEC 80003, Quantities and their units to be used in physiology, 6 parts

IEC/TC 25 has decided that the title for the ISO 80003 and IEC 80003 series should be "*Quantities and their units to be used in physiology*". ISO/TC 12 approves of this change (Resolution 13).

Mr. G r me reported on the progress of IEC 80003-1, *Modalities*.

Mr. Obrd  lek reported on the progress of ISO 80003-2, *Telebiometrics related to physics*. He asked how to proceed regarding the OID (Object IDentifier) column in the tables.

There was a lengthy discussion on OID. Mr. G r me explained how OIDs are created and the hierarchy. Mr. Gessner explained the practice in standardization committees for healthcare. ISO/TC 184/SC 4 is also involved. (Resolutions 14 and 15).

Mr. Obrd  lek asked about the views on duplication of definitions from the ISO 80000 and IEC 80000 series to the ISO 80003 and IEC 80003 series. He was advised not to duplicate definitions but rather give the symbol, OID and reference to the relevant part of ISO 80000 or IEC 80000.

Ms. Losada Gonzalez reported that the work on IEC 80003-4, *Telebiometrics related to biology* is progressing.

Mr. Brinkmann asked why this part is allocated to IEC and not ISO and the Chairman replied that IEC/TC 25 decided this at the plenary meeting in Tel Aviv in 2009 and that ISO/TC 12 has been notified but has not made any comments about this within the time limit for objections.

Mr. Gérôme reported on IEC 80003-5, *Telebiometrics related to culturology*, and IEC 80003-6, *Telebiometrics related to psychology*. There will be drafts for the IEC/TC 25 meetings in August 2010.

Mr. Gérôme explained that the word "culturology" is a Russian term defined as "integral study of human cultures as integral systems and their influence on human behaviour".

Mr. Jacobson asked if the order of the parts should be changed so that *Psychology* should come before *Culturology*. Mr. Gérôme replied that the present order had been the result of long and intense discussions in the WG.

7.3 ISO/CD 80003-3, Quantities and their units to be used in physiology - Part 3: Telebiometrics related to chemistry

The project leader Ms. Fylkner reported that the document is currently under CD ballot until 2010-06-12. Clause 11, quantities and units for remedies (in chemistry) has been supplied by Mr. Gessner.

The title was discussed and the present members of ISO/TC 12 recommended that the title be changed to *Quantities and their units to be used in physiology - Part 3: Telebiometrics related to chemistry*. The member bodies of ISO/TC 12 are encouraged to make this comment on the CD ballot.

It was noted that clause 10, *Quantities and units for signs observed in CALOR-IN and COLOR-OUT* is empty. It was decided to transfer this clause from ISO 80003-3 to ISO 80003-2..

The Secretary was asked to request ITU to comment on this document.

Mr. Gessner proposed that the unit symbol is the code and thus the OID column for the unit shall be deleted. OID codes for the quantities, however, shall remain.

Mr. Gérôme showed an excel sheet for the drafting of an annex listing the units used for different quantities. These units need not to be SI base units, but could also be derived units, and multiples and submultiples.

Mr. Brinkmann pointed out that the physical object or phenomena will put constraints to the use of units. If the object "table" is considered the units m, dm, cm, and mm could be considered but not km or nm.

The OID codes were discussed in much detail. A resolution effects that unit symbols are sufficient identifiers and OID codes are not necessary (Resolution 15) and the OID column for units shall be deleted in the tables. This decision shall be conveyed to WG 13 and IEC/TC 25/WG 6.

8 Review of liaisons

8.1 ISO committees in liaison

The Chairman gave the background to the internal liaisons. ISO/TC 12 noted the list of internal decisions and decided not to delete any of the present liaisons (Resolution 16). P-members have access to information from ISO/TCs in liaison.

Mr Brinkmann proposed that ISO/TC 12 should request liaison with ISO/TC 8/SC 6, *Ships and marine technology - Navigation and ship operations*.

Mr Gessner proposed that ISO/TC 12 should request liaison with ISO/TC 215, *Health informatics*.

It was decided that ISO/TC 12 should seek liaison with ISO/TC 8/SC 6 and ISO/TC 215 (Resolution 16).

8.2 IEC committees in liaison

The Chairman reported on the liaison with IEC/TC 25, *Quantities and units*. The Chairman is also Chairman of IEC/TC 25.

Mr. Schwob proposed that ISO/TC 12 should form liaison with IEC/TC 1, *Terminology*. It was decided that ISO/TC 12 should seek liaison with IEC/TC 1 (Resolution 17).

8.3 Organizations in liaison

The Chairman reported on the organizations in liaison.

Ms. Pellaux pointed out that IAEA and OIML do not have liaison officers for ISO/TC 12 registered in the ISO Global Directory. Ms. Pellaux will contact OIML and the Secretary will contact IAEA.

Ms. Pellaux pointed out that B liaisons can only be maintained with intergovernmental organizations. The Secretary was asked to investigate whether IMO should be listed as A or B liaison.

Mr. G  r  me proposed that ISO/TC 12 should request liaison with ICO, International Customs Organization, ICAO, International Civil Aviation Organization and ILO, International Labour Organization (resolution 18). This proposal was approved.

8.4 Request for liaison from OASIS

Mr. Thompson red the presentation from OASIS. There was a discussion how to proceed. The present members of ISO/TC 12 recommend the members of ISO/TC 12 to accept the request for liaison. The secretary was tasked to inform OASIS on the discussions.

9 Items for future work

Mr. Gessner raised the question about the standardization of **data coding for quantities and units**. In the past the printed document was the data carrier and ISO/TC 12 has put a great effort in standardizing the symbols.

The Chairman presented the Swedish standard for *Diagrams and tables*. Mr. Brinkman showed the corresponding DIN standard. The committee encouraged Sweden to make a new work item proposal on this topic.

ISO published new **ITSIG rules** in May 2009. ISO/TC 12 asked its secretariat to make excerpts of the relevant pages and circulate to the committee together with a draft letter containing comments.

The Chairman presented a Swedish standard for *vector quantities*.

Mr. Brinkmann presented DIN 1338, *Writing and typesetting of formulae*. One problem is that the leading software producer does not comply to ISO rules.

10 Follow-up of the ISO 80000 and IEC 80000 series

The Chairman summed up the discussions on the different parts of ISO 80000 and IEC 80000 series.

11 Requirements concerning a subsequent meeting

The Chairman proposed that ISO/TC 12 should enquire the possibility to hold a joint meeting together with IEC/TC 25, possibly in Norway in the autumn of 2012.

12 Any other business

The ISO program manager for ISO/TC 12, Ms. Pellaux, reported that ISO plans to print a **ISO/IEC Handbook for quantities and units** during the second half of 2010. She put the question whether the new edition should be published as hard copy, CD or database.

Mr. Thompson requested that the **ISO SI Guide** should be free of charge on Internet and that it should also give the references to the relevant standards. Ms. Pellaux replied that she would forward this comment to the person in charge at ISO/CS. She also pointed out that **ISO concepts database** gives the definitions from the standards by ISO/TC 1,2 but that they should be reviewed by the secretariat of ISO/TC 12.

Mr. Brinkmann gave a Powerpoint presentation on a project by Siemens, doc. N 858.

Mr. G  r  me gave a presentation on General Telebiometry, doc. N 859.

Mr. Jacobson asked about the relation between IEC 80000-14, and ISO 80003 and IEC 80003 and the reason why parts 4, Biology, Part 5, *Culturology* and Part 6, *Psychology* are developed by IEC.

13 Approval of resolutions

20 resolutions were approved (see ISO/TC 12, Doc. 860).

14 Closure of the meeting

The Chairman expressed the appreciation of ISO/TC 12 to UNMZ for hosting this meeting and especially to Mr. Obrd  z  lek. He then closed the meeting at 14:45 on Tuesday 13 April 2010.